



Electrical THz pulses for electronic transport characterization

P.-B. Vigneron, Nathan Roussel, Raphaël Pederiva, Thomas Vasselon, Clément Geffroy, Kevin Bredillet, Giorgos Georgiou, Christopher Bäuerle, Jean-François Roux

► To cite this version:

P.-B. Vigneron, Nathan Roussel, Raphaël Pederiva, Thomas Vasselon, Clément Geffroy, et al.. Electrical THz pulses for electronic transport characterization. ITQW 2025, Raffaele Colombelli; Jean-Michel Manceau, Sep 2025, Saint Malo, France. <hal-05293904>

HAL Id: hal-05293904

<https://hal.science/hal-05293904v1>

Submitted on 2 Oct 2025

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



HAL Authorization

Electrical THz pulses for electronic transport characterization

Pierre Baptiste Vigneron¹, Nathan Roussel¹, Raphael Pederiva¹, Thomas Vasselon², Clément Geffroy², Kevin Bredillet¹, Giorgos Georgiou³, Christopher Bäuerle², Jean-François Roux¹

¹ CROMA, CNRS, UMR 5130, University Grenoble Alpes, Univ. Savoie Mont-Blanc, Grenoble INP, Grenoble, France

² Institut NEEL, Univ. Grenoble Alpes, CNRS, Grenoble INP, Grenoble, France

³ James Watt School of Engineering, Electronics and Nanoscale Engineering, University of Glasgow, Glasgow, U. K.

*Contact Email: pierre-baptiste.vigneron@univ-smb.fr

Short Abstract Electrical THz pulses can be generated by using ultrafast photoswitches illuminated by optical laser pulses. We have demonstrated the excitation of an electronic plasma, and we investigate techniques to shape the THz pulse with durations ranging from 2.3 ps to 15 ps. These results pave the way for the design of a system to generate guided THz pulses with arbitrary amplitude and shape, to be used to excite and control electronic qubits.

1. Introduction

Integrated optoelectronic circuits using ultrafast photoswitches (PSW) for picosecond pulse generation and detection enable the study of the THz frequency response of a 2D material such as graphene [1] or two-dimensional electron gas (2DEG) [2] in which electron wavepackets can be generated. These wave packets can be used to generate electronic qubits that interact strongly through Coulombic interaction [3]. Recently, using 30 ps pulses generated by an electronic arbitrary waveform generator, Ouacel *et al.* paved the way for the dynamic control of such electronic qubits [4]. However, the use of even shorter pulses would enable to excite single-electron qubits and increase the number of flying qubits that can be manipulated within a single processing unit. So, it is necessary to design a circuit that delivers both short pulses of 1 to 2 ps to excite the electronic qubits, and longer pulses of 5 to 10 ps to control the gates dynamically. For this, we intend to use THz optoelectronic circuits driven by NIR laser pulses with different shapes and durations.

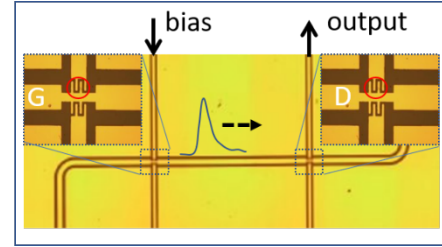


Fig. 1. Picture of the integrated optoelectronic THz circuit. The ps pulses are guided by a central line. They are generated by a pair of biased PSW (identified by G) illuminated by laser pulses with variable duration and shape. After 1mm of propagation, the THz pulses are detected by a dual pair of PSW (identified by D) triggered by 60fs delayed laser pulses. The output signal is the current delivered by one of the detection PSW.

2. Results

The typical THz optoelectronic circuit is described in Fig. 1. It is based on a gold integrated circuit deposited onto a 300 nm thick layer of LTG-GaAs with ultrafast properties. The layer is grown on a Si-GaAs wafer topped by a Bragg grating that optimizes the laser pulse absorption in the ultrafast layer. The carrier electron lifetime of LTG-GaAs is 500 fs as measured by time domain optical reflectometry. The THz circuit consists in a central coplanar waveguide (CPW) and four perpendicular access lines with interdigitated PSW designed at their ends. One pair of face-to-face PSW (named G) is used to generate the electrical pulses whereas the other pair of PSW (named D) is used to sample the ps electrical pulse after a few-mm of propagation.

A first challenge is to couple and characterize the voltage pulse propagating on a CPW in the 2DEG to generate a single electron wavepacket (SEW). The coupling is performed with an Ohmic contact on top of

the 2DEG connected to the PSW of generation (G) by a CPW (see inset Fig.2). After propagation in the 2DEG, the SEW is out-coupled with a second Ohmic contact to the CPW. A second PSW is used for detection (D). The two Ohmic contacts are separated by 150 μm . In the figure 2, we plot the time response of the 2DEG plasma excited by a THz electrical pulse. The cavity confines the motion of the 2D plasmon between the two Ohmic contacts, forming a Fabry-Pérot resonance. The spectrum of this time response (inset of figure 2) exhibits a dip of transmission at $f = 88.5$ GHz. From these observations, we calculated the pulse velocity at $v_{2\text{DEG}} = 2.12 \cdot 10^7 \text{ m.s}^{-1}$. In contrast, the pulse velocity in the CPW is about $v_{\text{CPW}} = 1.2 \cdot 10^8 \text{ m.s}^{-1}$.

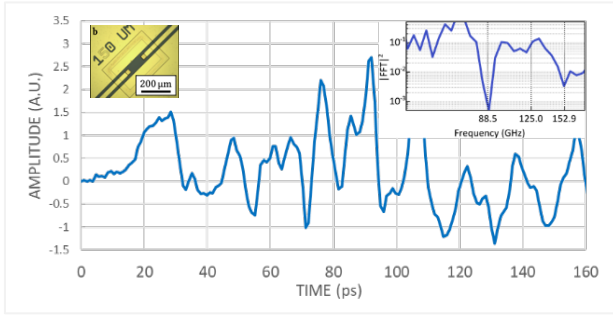


Figure 2: Time response of a 2DEG plasma excited by a THz electrical pulse at 4 K. Inset: left) image of the CPW on both side of a 2DEG sheet, right) FFT of the signal, highlighting the first resonance frequency of the 2DEG system around 88.5 GHz.

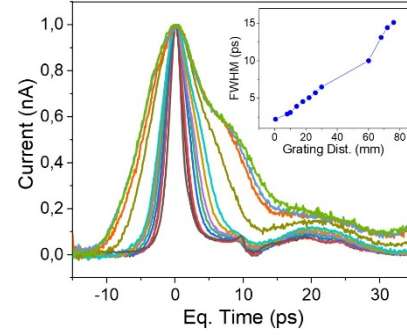


Figure 3: Normalized THz pulses generated using laser pulses dispersed by a pair of gratings separated by D ranging from 0.6 to 78 mm. The inset shows evolution of the THz pulse width (FWHM) vs the distance D .

A second stake of this project is to control the propagation of the SEW along the 2DEG with electrical gates. To ensure adequate control, the opening time of these gates should be tunable for 1 to 20 ps. The solution we propose is to trigger the gates with an electronic THz pulse with arbitrary duration. We propose an optical approach to this problem; by using a pair of optical gratings in a double-pass arrangement to engineer the NIR laser pulse dispersion. The dispersion is controlled by changing the distance D between the two gratings. We have plotted in Fig. 3 all the normalized data obtained for D ranging from 0.6 to 78 mm. The FWHM of the detected THz pulses consequently varies from 2.3 to 15 ps in a quasi-linear manner, as shown in the inset of Fig. 3. For the longer pulses, we observe a change of the temporal shape with the birth of a shoulder around 7 ps. This deformation may be due to unwanted non-CPW propagation modes of the THz waveguide or to higher-order chromatic dispersion in the pair of gratings. Soon, using a more advanced dispersive setup based on diffraction gratings and prisms could mitigate the effect of higher-order dispersion on the shape of the stretched optical pulse [5]. These initial results constitute a first step toward the design of a system for generating guided THz pulses with controllable amplitude and shape through optical pulse shaping techniques.

References

- [1]. Yoshioka, K., Bernard, G., Wakamura, T. *et al.*, "On-chip transfer of ultrashort graphene plasmon wave packets using terahertz electronics," *Nat Electron* 7, 537–544 (2024). DOI: [10.1038/s41928-024-01197-x](https://doi.org/10.1038/s41928-024-01197-x)
- [2]. J. Wu, A. Mayorov, C. Wood, *et al.*, "Excitation, detection and electrostatic manipulation of terahertz-frequency range plasmons in a 2-DES," *Sci Rep.* 5, 15420. DOI: [10.1038/srep15420](https://doi.org/10.1038/srep15420) (2015).
- [3]. C. Bäuerle, C. Glatli, T. Meunier *et al.*, "Coherent control of single electrons: a review of current progress" *Rep. Prog. Phys.* 81 056503 (2018). DOI [10.1088/1361-6633/aaa98a](https://doi.org/10.1088/1361-6633/aaa98a)
- [4]. S. Ouacel, L. Mazzella, T. Kloss *et al.*, "Electronic interferometry with ultrashort plasmonic pulses." [10.48550/arXiv.2408.13025](https://arxiv.org/abs/10.48550/arXiv.2408.13025) (2024). DOI: [10.48550/arXiv.2408.13025](https://doi.org/10.48550/arXiv.2408.13025)
- [5]. N. Forget, V. Crozatier and P. Tournois, "Transmission Bragg-grating grisms for pulse compression," *Appl. Phys. B* 109, 121–125 (2012). DOI: [10.1007/s00340-012-5126-2](https://doi.org/10.1007/s00340-012-5126-2)